

Topics:

- ☐ Difference between Average and Instantaneous Rates of Change.
- ☐ Approximating the Derivative.
- ☐ Given a graph of data points, answer questions relating to rates of change.
- ☐ Given $f(x)$, find $f(x + \Delta x)$.
- ☐ Given $f(x)$, find $f'(x)$.
- ☐ Given $f(x)$ and $x = c$, find the tangent line of $f(x)$ at c .
- ☐ Use the tangent line of $f(x)$ to approximate $f(x)$.
- ☐ Find derivatives using the derivative rules
 - product rule
 - quotient rule
 - chain rule
 - implicit differentiation
- ☐ Given $f(x)$, find the critical points of $f(x)$.
- ☐ Given $f(x)$, find the intervals of increase and decrease of $f(x)$.
- ☐ Given $f(x)$, find the intervals of concavity of $f(x)$.
- ☐ Given $f(x)$, find the local maximum and minimum of $f(x)$.
- ☐ Given $f(x)$, find the absolute maximum and minimum of $f(x)$.

Problems:

1. Let $f(x) = x^2 - x$. Answer the following questions:

(a) Find and simplify $f(x + \Delta x)$

(b) Find and simplify $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

(c) Use the limit definition of the derivative to find $f'(x)$

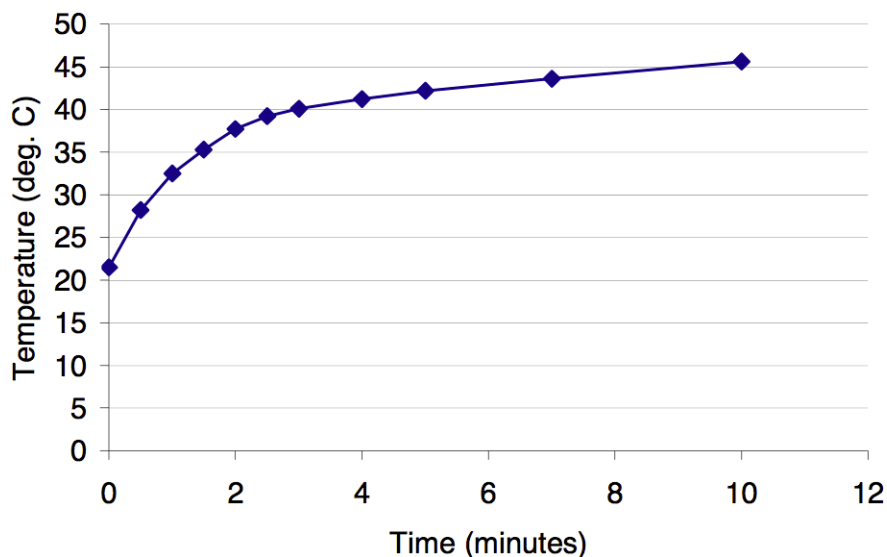
(d) Find the slope of the tangent line of $f(x)$ at $x = 1$

(e) Find a point on the tangent line of $f(x)$ at $x = 1$

(f) Find the equation of the tangent line of $f(x)$ at $x = 1$

(g) Use part (f) to approximate $f(1.05)$

2. Suppose an object is placed in an industrial oven. The graph below represents the temperature, T , of an object over time, t . Use this graph to answer the questions that follow:



- (a) What was approximate temperature of the object before it was placed in the oven?
- (b) What is the approximate average rate of change of the temperature during the first 4 minutes. (*include units*)
- (c) Suppose $T'(3) = .625$ °C/min. What is the equation of the tangent line of $T(t)$ at $t = 3$
- (d) Consider the quantities below and list them in increasing order.

(I) Instantaneous Rate of change of T at $t = 4$

(II) Average Rate of change of T from $t = 0$ to $t = 4$

(III) Concavity of T at $t = 4$

3. Using the derivative rules, find the derivatives of the following functions:

(a) $f(x) = x^5 - x$

(g) $f(x) = 7^2$

(b) $g(x) = 2^x$

(h) $f(x) = \ln x + e^x$

(c) $f(r) = r \ln r$

(i) $f(x) = \frac{x}{x-1}$

(d) $f(x) = \left(\frac{x}{x-1} \right)^3$

(j) $h(x) = \frac{\sqrt{x^2 - 1}}{2}$

(e) $f(t) = e^{t^{19}}$

(k) $P(x) = \ln(2x - e^x)$

(f) $f(x) = \frac{10^x}{\ln(3-x)}$

4. Given each of the equations below, compute $\frac{dy}{dx}$.

(a) $x^3 = y^4 - x$

(c) $xy = (y - x)^2$

(b) $g(x) = 2^x$

(d) $ye^x = \ln y + 1$

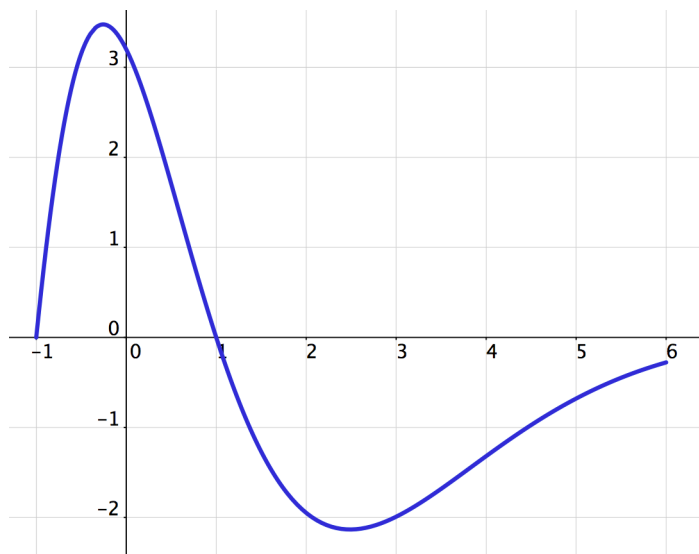
5. Find the equation of the tangent line of the curve $2y = x + y^2$ at $(1, 1)$.

6. Let $E(t)$ be the number of eggs a farmer produces and sells on day t , and $C(t)$ is the cost per carton (which also changes daily). If the farmer starts selling eggs on May 1st, then...

(a) What is the expression that represents the farmer's total earnings (revenue) on May 10th?

(b) What is the expression that best describes the farmer's rate of revenue increase on May 10?

7. Below is the graph of some function, $g(x)$ on the interval $[-1, 6]$. Using the graph, answer the questions that follow. **Be prepared to explain your answer.**



(a) Fill in each box with either $<$, $>$, or $=$.

$\succ g(0) \quad \square \quad 0$

$\succ g'(0) \quad \square \quad 0$

$\succ g''(0) \quad \square \quad 0$

$\succ g(2) \quad \square \quad 0$

$\succ g'(2) \quad \square \quad 0$

$\succ g''(2) \quad \square \quad 0$

$\succ g(1) \quad \square \quad g'(1)$

$\succ g'(2.5) \quad \square \quad g''(2.5)$

$\succ g''(2) \quad \square \quad g''(5)$

$\succ$ average rate of change from -1 to $2 \quad \square \quad$ average rate of change from -1 to 3

(b) List the (approximate) critical points of $g(x)$

(c) Give the (approximate) interval(s) where $g(x)$ is increasing.

(d) Give the (approximate) interval(s) where $g(x)$ is decreasing.

8. Consider the function $f(x) = x^3 - 3x + 10$. Find the following

(a) Critical points of $f(x)$.

(b) Find the absolute maximum and minimum on the interval $[-2, 4]$.

(c) Use the first derivative test to classify the critical points as either local max, local min, or neither.